

A Fluorine-Free Binder with Organic–Inorganic Crosslinked Networks Enabling Structural Stability of Ni-Rich Layered Cathodes in Lithium-Ion Batteries

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Although the high-energy Ni-rich layered cathodes suffer from undesirable surface reactions with the electrolyte, the polyvinylidene fluoride (PVDF) binder has a limitation on surface stabilization because of its weak affinity and low adhesion/cohesion. Here, it is demonstrated that the novel fluorine-free and hydroxyl-rich siloxane nanohybrid (SNH) binder can enhance the electrochemical performances of $\text{LiNi}_{0.8}\text{Mn}_{0.1}\text{Co}_{0.1}\text{O}_2$ cathode (NCM811) via successful surface stabilization. The high silanol content in the SNH binder enhances the affinity to both NCM811 and conductive agent, facilitating uniform electron/ion pathways with high mass loading, improved shear thinning, and superior mechanical properties. Moreover, the fluorine-free organic-inorganic hybrid structure prevents the dissolution of transition metals, active material structural changes, and electrolyte interaction, leading to greatly enhanced cyclability of the SNH-based NCM811 electrode ($\approx 81.9\%$ in half-cell; $\approx 87.82\%$ in full-cell after 200 cycles) compared to PVDF-based NCM811 electrode ($\approx 58.8\%$ in half-cell; $\approx 61.24\%$ in full-cell after 200 cycles). Various analyses also indicate that the application of the fluorine-free SNH binder successfully stabilizes both the surface and bulk structure of the NCM811 cathode during charge/discharge. The binder design represents a straightforward yet highly effective approach to achieving remarkably prolonged cyclability in lithium-ion batteries, surpassing the performance of other fluorine-based or polymer-based binders.

1. Introduction

With the implementation of eco-friendly policies worldwide, the demand for exhaust-free electric vehicles (EVs) is increasing.^[1,2] Lithium-ion batteries (LIBs) have high energy density, which is a factor in their rapid market growth.^[3–5] To develop a LIBs that can travel a longer distance on a single charge, it is essential to understand the materials of each component. High-capacity Ni-rich cathode materials like $\text{LiNi}_{0.8}\text{Mn}_{0.1}\text{Co}_{0.1}\text{O}_2$ (NCM811) have a practical capacity $> 200 \text{ mA h g}^{-1}$, so they are increasingly used.^[6,7]

However, fabrication of cathode electrodes that have sufficient electrochemical stability for use in high-energy-density LIBs, presents an ongoing challenge.^[8,9] Key difficulties include non-uniform conductive pathways for electrons and ions,^[10,11] poor adhesion to electrodes,^[12,13] and dissolution of transition metal (TM) ion from cathode materials.^[10,14] Particularly, when the electrolytes include carbonates, the Ni^{3+} or Ni^{4+} reacts with

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them and is reduced to Ni^{2+} , which can migrate from the TM layer to the Li^+ layer within the crystal lattice.^[15,16] This is because the lower ionic radii of Ni^{2+} have a smaller ionic radius ($\approx 0.69 \text{ \AA}$) than Li^+ ($\approx 0.76 \text{ \AA}$) so substitution of Ni^{2+} for Li^+ in the lattice can distort it and thus cause internal cracking in particles and irreversible formation of non-conductive NiO -based rock salt phase.^[17,18] These side reactions degrade the cathode. One way to avoid this problem is to modify the cathode's components, such as binders.^[13]

Polyvinylidene fluoride (PVDF) is commonly used as a cathode binder material due to its oxidation resistance against oxide cathode materials, and its electrochemical stability in the operational conditions of LIBs.^[19,20] However, several limitations hinder the development of stable cathode electrodes. These include high cost, low mechanical ductility, weak affinity to active or conductive materials, low solid content ($<10\%$) in solution leading to difficulty in achieving thick electrodes, and use of toxic solvents such as *N*-methyl-2-pyrrolidone (NMP), resulting in inevitable swelling and pulverization of the active electrode materials during repeated cycles.^[14,21,22] Additionally, European Union regulations will legally restrict perfluoroalkyl substances, which include PVDF, due to their potential to cause environmental pollution. Therefore, ecologically benign binder materials without fluorine must be developed.^[23] Various studies have focused on developing alternatives that use organic chemicals or biomaterials such as acrylate-functionalized polymers,^[13,24] oligoethers,^[25] and organic-based elastic binders.^[26–28] These efforts have improved mechanical properties and electrochemical stability in battery cells by either replacing PVDF binder materials or by designing molecular structures. However, binders that use organic materials have limited electrochemical stability during long-term cycling and rate capability tests. The materials have low solid content, so fabrication of thick electrodes is a difficult task.^[29,30] Also, evaporating toxic solvents during the fabrication of the electrodes and complex processes for binder development hinder the commercialization of those binders-applied LIBs.^[31] To avoid these problems, new types of binder materials must be developed.

Herein, we introduce a novel cathode binder material based on fluorine-free and resilient organic–inorganic hybrid type binder for LIBs that use NCM811 cathodes. The binder uses cycloaliphatic epoxy-based siloxane nanohybrid materials (SNHs), which are synthesized using a simple sol-gel synthesis approach. Molecular-scale chemical hybridization between organic functional groups (cycloaliphatic epoxy and silanol) and an inorganic network (Si-O-Si bond) results in electrochemical stability and mechanical strength. Key advantages of SNH compared to other fluorine-free binder materials are high stability originating from strong bonding energy of siloxane backbone compared to the carbon-based backbone (bonding energy: $\text{Si-O} = 452 \text{ kJ mol}^{-1}$, $\text{C-C} = 346 \text{ kJ mol}^{-1}$), high solid content, and simple but effective synthesis method.^[32] The SNH binder has stronger adhesion and cohesion than conventional PVDF binders. The high content of silanol groups in the SNH increases its affinity to active materials and conductive agents, and thereby effectively prevents dissolution of TM ions during battery operation. Thus, this SNH binder can suppress the severe structural changes that occur during charge/discharge cycles. Through the combined studies using first-principles calculations and various experiments, it was confirmed that SNH binder interacts strongly with the

surface of NCM811 particles. It also but also achieves the uniform NCM811 particle coating through its homogeneous distribution within the electrode. The NCM811 electrode that uses the SNH binder showed an outstanding cyclability with capacity retention of $\approx 81.9\%$ after long-term cycling, whereas the NCM811 electrode that uses PVDF binder achieved only $\approx 58.8\%$ under the same conditions. Also, the SNH binder improves the electrochemical characteristics of the full-cell compared to full-cells that use PVDF binders.

2. Results and Discussion

2.1. The Characterization of SNH Binder

The SNH binder was synthesized by hydrolytic sol-gel reaction (hydrolysis and condensation) of (2-(3, 4-epoxycyclohexyl)ethyl)trimethoxysilane (ECTS) with tetraethyl orthosilicate (TEOS) (Figure S1, Supporting Information) under base-catalyst.^[14,30] The binder has a high affinity to NCM811, so it covers the NCM811 particles uniformly without inducing them to aggregate (Figure 1A). Synthesis of SNH binder entails two kinds of reaction: 1) hydrolysis that generates hydrolyzed groups (i.e., Si-OH) that increase affinity to active material and conductive agent; and 2) condensation that yields robust siloxane networks that improve electrochemical stability. These two kinds of reaction are essential to achieve highly stable operation of NCM811-based LIBs during charge/discharge.

For sol-gel processes, many reactive species can be used to generate hydrolysis or condensation (three methoxy groups in ECTS and four ethoxy groups in TEOS). Accordingly, even if highly condensed siloxane networks form under the base-catalyst, the synthesized SNH should retain many hydrolyzed species (i.e., $-\text{OH}$).^[14] To confirm this possibility, we investigated the chemical structure of the synthesized SNH binder. The ^{29}Si -nuclear magnetic resonance (^{29}Si -NMR) spectrum indicated the formation of siloxane bonds, represented by the terms T^n (from ECTS) and Q^n (from TEOS), where superscript n denotes the number of siloxane bonds of the Si atoms (Figure 1B). In ^{29}Si -NMR result, the T^3 species peaks were mainly detected; this result means that alkoxide groups in ECTS had been fully converted to siloxane networks with a branched network structure and high degree of siloxane bond formation ($\approx 94.23\%$) (i.e., highly condensed siloxane networks formed) (Table S1, Supporting Information) that can be also confirmed by changes of Fourier-transformed infrared (FT-IR) spectra by comparing before (i.e., silane mixture) and after (i.e., SNH resin) sol-gel reaction (Figure S2, Supporting Information).^[33] In addition, the broadened peaks of T^2 and Q^0 species peaks, and distinct Q^1 species peaks were confirmed; this result indicates the generation of a hydrolyzed silanol group (from alkoxide to hydroxyl); FT-IR spectra (Figure 1C) confirmed this result. The generation of band at $3150\text{--}3700 \text{ cm}^{-1}$, which corresponds to $-\text{OH}$ stretching, and the decreasing intensity of the band at 870 cm^{-1} , which indicates the epoxy group, were observed. These results indicate that cured SNH binder retained hydroxyl groups.

The molecules in the SNH binder have a diverse molecular weight (until $10\,000 \text{ dal}$), as confirmed by matrix-assisted laser desorption/ionization (MALDI-TOF) analysis; this result means that hydrolyzed species (i.e., hydrolyzed ECTS and TEOS)

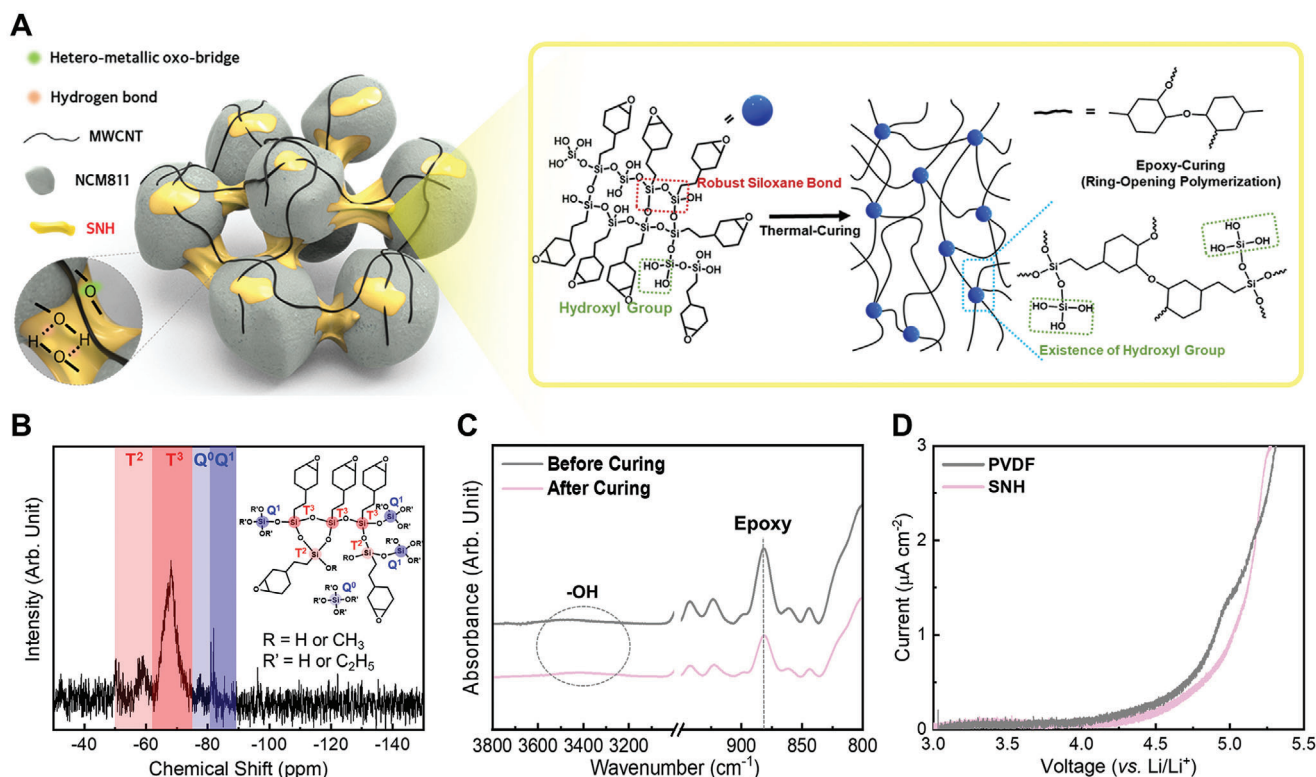


Figure 1. A) Schematic illustration of structure of SNH-applied cathode. B) ^{29}Si -NMR analysis of SNH resin (inset: chemical formula of SNH binder). C) FT-IR spectra before and after curing of SNH. Curves offset vertically for clarity. D) Results of LSV of PVDF (gray) and SNH binders (pink).

remained, and supports the results of FT-IR and ^{29}Si -NMR (Figure S3, Supporting Information). These results indicate that the SNH binder with highly condensed siloxane networks has undergone partial silanol substitution because of the hydrolyzation of ECTS and TEOS. These networks have a high affinity with NCM811, and therefore can homogeneously cover it.^[14]

To quantify the interaction between conductive agents (carbon nanotubes, CNTs) and SNH binder, we conducted Raman mapping analysis to compare pristine CNT film, CNT/PVDF mixture, and CNT/SNH mixture. The SNH binder interacted more strongly than PVDF with CNTs (Figure S4, Supporting Information).^[34]

Comparison of linear sweep voltammetry (LSV) profiles of SNH and PVDF binders demonstrated that SNH binder is more stable against electrochemical oxidation than PVDF binder with a higher onset voltage (onset voltage: SNH = 4.9 V; PVDF = 4.6 V) (Figure 1D).^[35] These results indicate that the SNH binder is successfully prepared by molecular hybridization of functional organic groups and robust inorganic networks; this process yields a high content of partially substituted silanol groups, which give electrochemical stability and a high affinity with active material and conductive agent. Furthermore, SNH binder has higher thermal stability than PVDF binder in which SNH binder exhibited a similar five weight percent thermal decomposition temperature ($T_{\text{d5wt\%}}$) and higher residue ($T_{\text{d5wt\%}}$ = 413 °C; residue = 42%) compared to PVDF ($T_{\text{d5wt\%}}$ = 420 °C; residue = 22%), supporting robustness of SNH binder (Figure S5, Supporting Information).

2.2. Properties of SNH Binder for Preparation of Electrode

To investigate the adhesion property of the SNH binder for the preparation of the electrode, we fabricated the SNH-based NCM811 slurry with the CNTs. The PVDF-based NCM811 slurry was also fabricated for comparison. The SNH slurry had higher shear-thinning behavior and coated the electrode more uniformly than the PVDF slurry; the difference is attributed to the higher dispersion of NCM811 with SNH binder than of PVDF binder (Figure 2A).^[36]

To quantify the adhesiveness of SNH and PVDF slurries, 180° peeling tests were conducted (Figure 2B; Figure S6, Supporting Information). The slurries had similar adhesion force (SNH slurry = 19.78 gf/20 mm; PVDF slurry = 19.65 gf/20 mm), so both had strong binding ability to the current collector. However, the cohesion of the SNH slurry (202 gf/12 mm) was higher than that of the PVDF slurry (120 gf/12 mm), so the SNH slurry had higher contact strength between layers of NCM811 than did the PVDF slurry. These results indicate that NCM811 particles are effectively wrapped with CNTs/SNH that have partial silanol substituents (Figure 2c; Figure S7, Supporting Information).

The SNH slurry had a lower complex modulus (17.92 Pa) than the PVDF slurry (42.55 Pa); i.e., the SNH slurry was more elastic than the PVDF slurry in the steady state (Figure S8, Supporting Information). Therefore, the SNH binder with high affinity to NCM811 cathode has better coating processability than the PVDF binder.

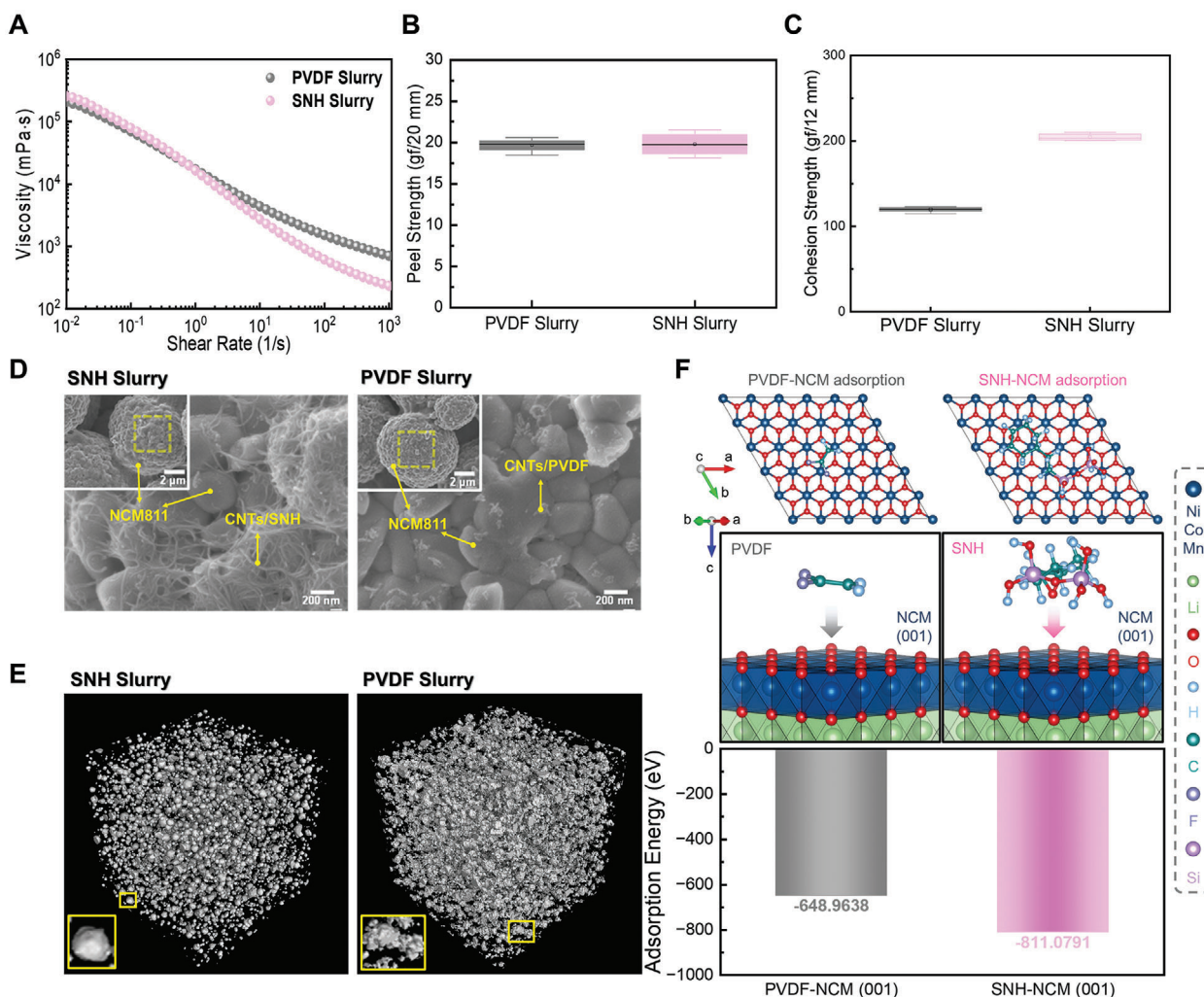


Figure 2. A) Rheological property of PVDF (gray) and SNH (pink) slurries. Mechanical properties of slurries coated with PVDF (gray) and SNH (pink) (B: peel strength, C: Cohesion strength). Morphology of SNH and PVDF slurries (D: SEM, E: X-ray CT). F) Calculated adsorption energies of each binder, and optimized geometries of PVDF and SNH binders on (001) surface of NCM.

We compared the morphology of both binders-applied slurries by using scanning electron microscope (SEM) and X-ray computed tomography (CT). The SNH binder uniformly covered CNT networks on NCM811 particles, whereas the PVDF binder yielded aggregated CNT networks (Figure 2D). Detailed surface morphology of both binders-applied slurries can be directly confirmed in (insets, Figure 2D). NCM811 particles and CNTs were uniformly distributed in SNH slurry, but somewhat aggregated in PVDF slurry (Figure 2E).

The high affinity of SNH binder to active material and conductive agent was investigated theoretically using density-functional theory (DFT) calculation (Figure 2F). We calculated the theoretical adsorption energies E_{ad} between NCM811 and binders as

$$E_{ad} = E_{\text{NCM811 with binder}} - (E_{\text{NCM811}} + E_{\text{binder}}) \quad (1)$$

where $E_{\text{NCM811 with binder}}$ is the total energy of the surface (001) of NCM811 that adsorbed the binder (i.e., SNH or PVDF), E_{NCM811} is the total energy of the pristine (001) surface of NCM811, and

E_{binder} is the total energy of the binder materials. Calculations yielded $E_{ad} = 4.95$ eV for SNH and $E_{ad} = 4.72$ eV for PVDF; the difference indicates that SNH has a higher affinity than PVDF to NCM811, and confirms the previous rheological, mechanical, and morphological analyses.

These results indicate that the developed SNH binder can facilitate uniform distribution of active materials and conductive agents, and thereby induce high mass loading and stability of LIBs. Furthermore, the excellent processability of SNH binder compared to PVDF binder demonstrates that SNH shows promise for use as a binder for Ni-rich layered cathode materials.

2.3. Effect of SNH Binder on Electrochemical Characteristics

To investigate the effect of SNH binder on the electrochemical characteristics, we compared changes in capacity during the cycling of SNH electrodes to those of PVDF electrodes, both in coin-type half-cells (Figure 3A–C). After 200 charge/discharge

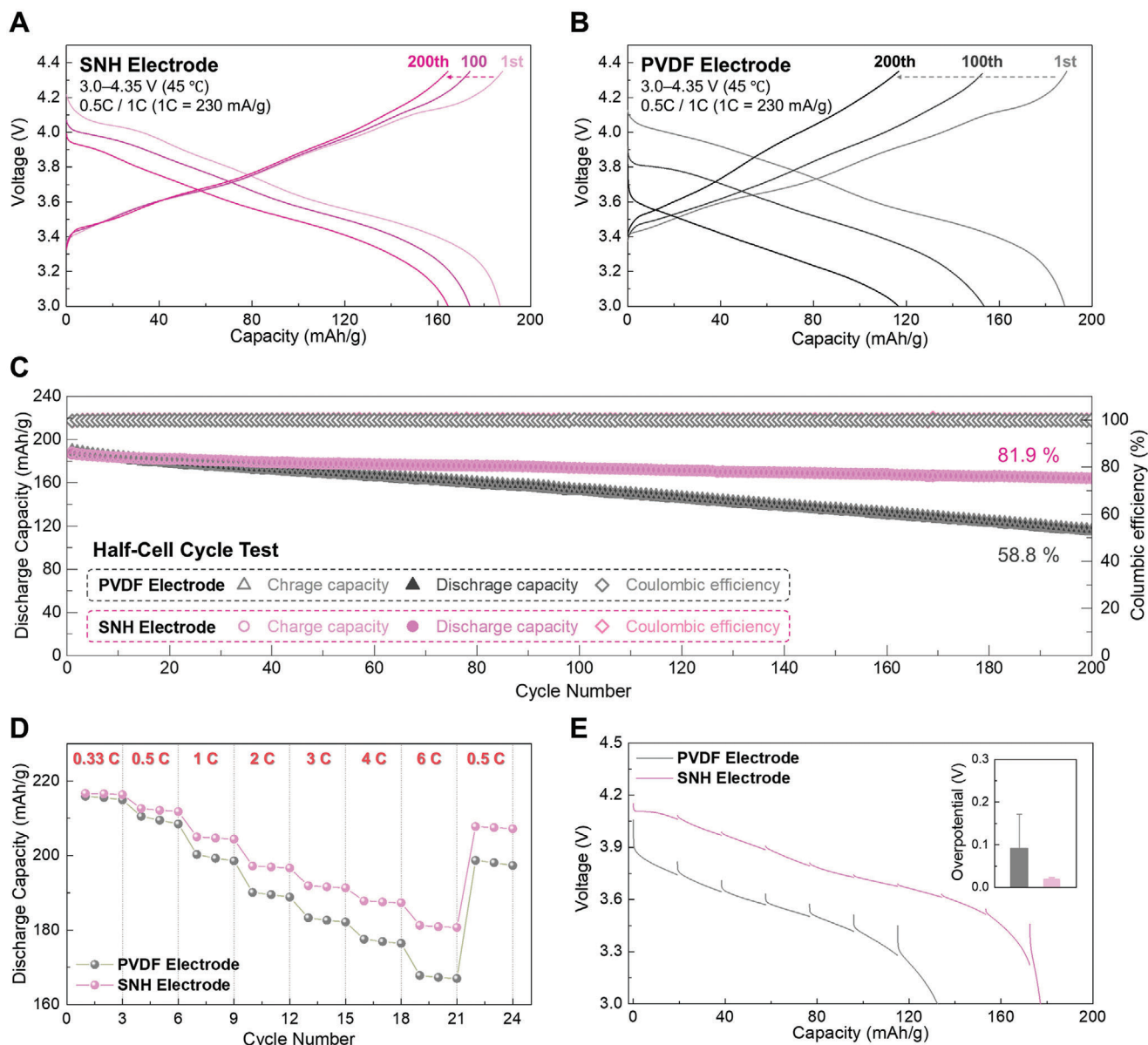


Figure 3. Characterization of electrochemical properties of SNH and PVDF electrodes. Charge/discharge curves at various current density of SNH (A) and PVDF electrodes (B). C) Cycle and D) rate decays of SNH electrodes (pink) and PVDF electrodes (gray). E) Overpotential of SNH (pink) and PVDF electrodes (gray).

cycles, the capacity retention was $\approx 81.9\%$ by the SNH electrode, by only $\approx 58.8\%$ by the PVDF electrode. Moreover, a comparison of the initial cycles verified that the SNH electrode had excellent Coulombic efficiency of $\approx 98.46\%$, whereas the PVDF electrode had $\approx 97.89\%$. The SNH electrode had good power capability; it retained $\approx 83.3\%$ of its original discharge capacity of 6C ($1\text{C} = 230\text{ mAh g}^{-1}$) at a discharge rate of 0.33C, whereas the PVDF electrode retained only $\approx 77.3\%$ (Figure 3D; Figure S9, Supporting Information). Galvanostatic intermittent titration technique (GITT) analyses confirmed this increased power capability of the SNH electrode. The overpotential of the SNH electrode (0.028 V) during charge/discharge was relatively small compared to that of the PVDF electrode (0.091 V). This result indicates that the SNH

binder has excellent electrochemical stability (Figure 3E). In addition, the Li-ion diffusion coefficient was calculated using the following equation:^[37]

$$D = \frac{4}{\pi \tau} \left(\frac{mV}{MS} \right)^2 \left(\frac{\Delta E_s}{\Delta E_t} \right)^2 \quad (2)$$

In the equation, D represents the diffusion coefficient of Li^+ , τ denotes the duration of application of constant current, m stands for the electrode's mass, V signifies the molar volume of the electrode, M corresponds to the molar mass of the electrode, S designates the surface area of contact between the electrode and electrolyte, ΔE_s represents the voltage change during a single step,

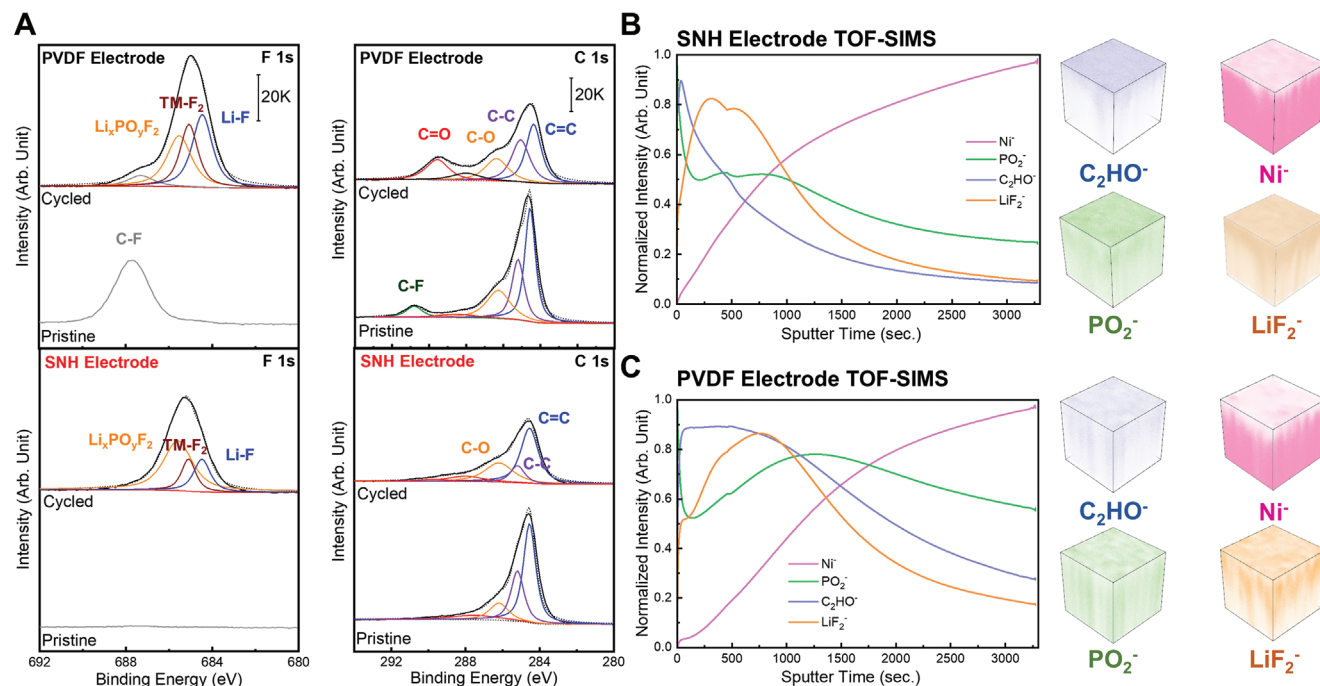


Figure 4. Elemental analyses of SNH and PVDF electrodes after charge/discharge cycle tests, A) XPS and B-C) TOF-SIMS with 3D mapping (B: SNH electrode, C: PVDF electrodes).

and ΔE_t reflects the cumulative voltage change under applied constant current conditions. As a result, the Li-ion diffusion coefficient was calculated to be $1.8 \times 10^{-8} \text{ cm}^2 \text{ s}^{-1}$ for the PVDF electrode and $3 \times 10^{-8} \text{ cm}^2 \text{ s}^{-1}$ for the SNH electrode. This difference implies that the SNH electrode facilitates Li^+ diffusion (Figure S10 and Table S2, Supporting Information).

The Nyquist plot comparing the SNH electrode to the PVDF electrode also provides evidence that the SNH binder had better electrochemical stability than the PVDF binder. The SNH electrode had lower interfacial resistance than the PVDF electrode (Figure S11, Supporting Information).

These results demonstrate that the SNH binder has a strong interfacial interaction that enables rigid packaging between the cathode active layer and the current collector.^[38] Also, the SNH slurry had a higher solid content (74.05 wt%) than the PVDF slurry (70.29 wt%) with the same composition. This high content can yield electrodes that have high mass loading (Table S3, Supporting Information).

2.4. Reduced Degradation of Ni-rich Layered Cathode by SNH Binder During Cycling

To identify why the SNH binder increases the electrochemical stability and power capability of the SNH electrode, we performed an X-ray photoelectron spectroscopy (XPS) analysis (Figure 4A; Figure S12, Supporting Information). The characteristic peaks at 684.5 and 685 eV corresponding to LiF and TM-F₂ (TM: Ni, Mn, Co) in F 1s spectra exhibited lower intensity in the SNH electrode than in the PVDF electrode. The formation of LiF and TM-F₂ results from the dissolution of TM ions from cathode materials by hydrofluoric acid in the electrolyte; these species re-

duce the cycle life of LIBs.^[18,39,40] Furthermore, in the F 1s XPS spectra, the SNH electrode showed a relatively lower intensity at 686 eV, corresponding to $\text{Li}_x\text{PO}_y\text{F}_z$, compared to the PVDF electrode; the difference implies that SNH effectively decreased the occurrence of side reactions with the electrolyte.^[41,42]

Comparison of C 1s spectra before and after cycling verified that use of SNH binder decreased both generation of unwanted fluorine-bearing byproducts and decomposition of electrolyte, whereas the spectrum of the PVDF electrode showed a strong C-F_x peak at $\approx 292 \text{ eV}$ that was not detected in the SNH electrode, and also showed strong peaks at 288–292 eV, which indicate C–O (from carbonate-based electrolyte or unreacted epoxide group in SNH) and C=O (from carbonate-based electrolyte), and are signs of electrolyte decomposition and unhappening of side reaction between unreacted epoxide group in SNH and carbonate-based electrolyte.

To further investigate the effect of the SNH binder on stabilizing the CEI-formation, we performed time-of-flight secondary ion mass spectrometry (TOF-SIMS) analyses. The outer CEI layer was identified by the location of maximum intensity of C_2HO^- . This layer was thinner in the cycled SNH electrode than in the cycled PVDF electrode (Figure 4B,C). The inner CEI layer was identified by the intensity of the insertion points between PO_2^- and Ni^+ in the cycled electrode.^[43] The locations of the outer and inner edges of the CEI its position, and the differences between their positions indicate its thickness. The CEI layer was lower and thinner in the cycled SNH electrode than in the cycled PVDF electrode. This result means that the SNH binder can induce stabilized formation of the CEI layer. The 3D TOF-SIMS maps of the cycled electrodes are consistent with this explanation.

Operando X-ray diffraction (XRD) analyses verified that the SNH binder can suppress the structural degradation of the

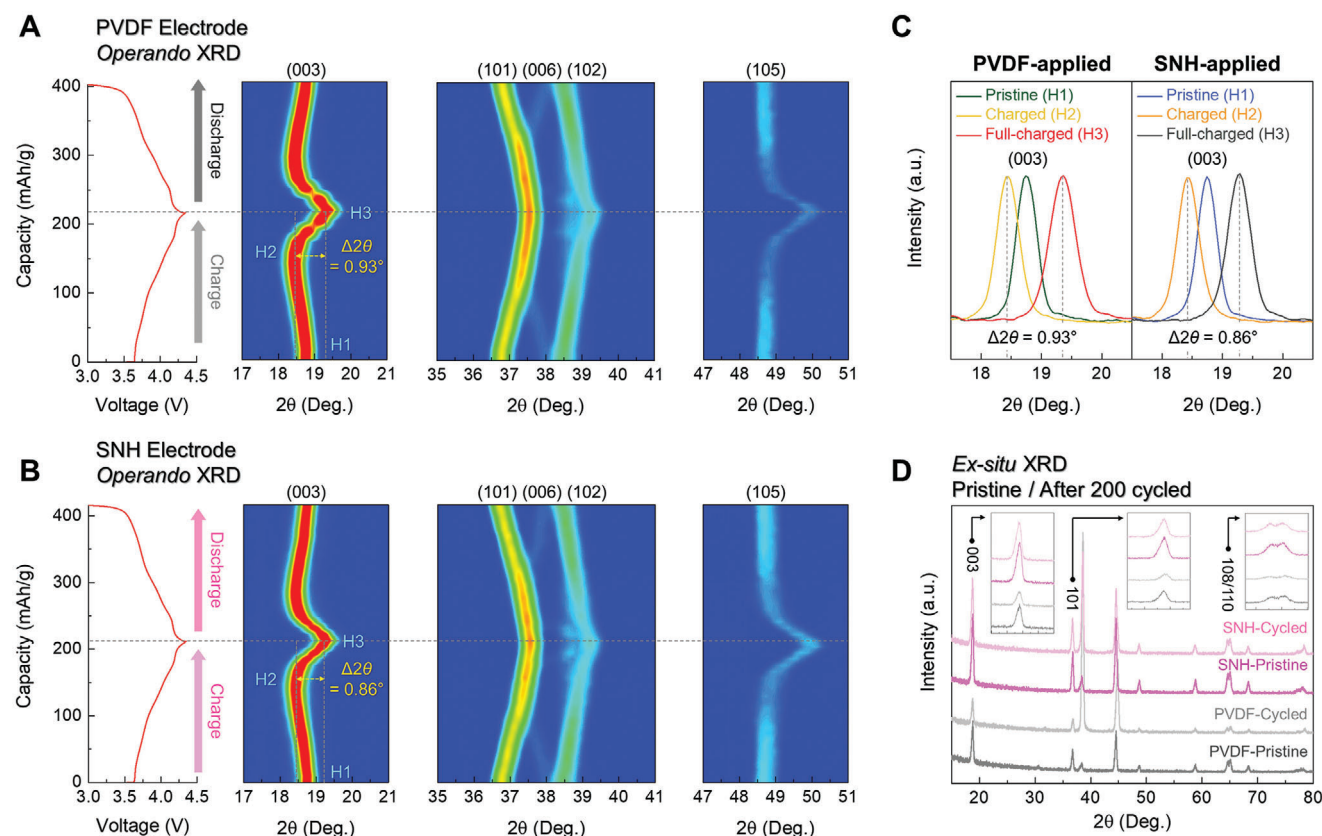


Figure 5. Operando XRD patterns with magnified views of NCM811 cathode using (A) PVDF binder and (B) SNH binder during charge/discharge. (C) Comparison of the (003) peak position during the H1, H2, and H3 phases transition in cathodes with PVDF and SNB binders, obtained using operando XRD results. (D) Ex-situ XRD analysis of electrodes with PVDF and SNH binders before and after cycling.

NCM811 cathode that occurs during charge/discharge. The XRD patterns (Figure 5A,B) of both samples exhibit the typical H1-H2-H3 phase transitions according to the state of charge (SOC).^[44,45] Specifically, the shift of the (003) peak toward lower angles during the charging process to 4.1 V (vs Li⁺/Li) indicates an expansion of the *c*-lattice and interlayer distance. These changes are associated with the H1-H2 phase transition in NCM811. When the LIB was fully charged to 4.35 V (vs Li⁺/Li), an H2-H3 phase reaction accompanied by a dramatic contraction of the *c*-lattice occurred, leading to a shift of the (003) peak to higher angles. Noticeably, during the H2-H3 transition, the shift of the angle of the (003) peak was smaller in the SNH electrode ($\Delta 2\theta = 0.86^\circ$) of than in the PVDF electrode ($\Delta 2\theta = 0.93^\circ$), even though both samples had almost the same SOC (Figure 5C). Moreover, Rietveld refinement based on the operando XRD patterns shows the total change in the lattice-parameters of the NCM811 cathode during charge/discharge (Figure S13, Supporting Information). In the PVDF electrode, the changes in the *a*- and *c*-lattice parameters were $\approx 2.0\%$ and 4.5% , respectively. In contrast, the SNH electrode exhibited suppressed structural changes during charge/discharge, with changes of $\approx 1.9\%$ and 4.0% in the *a*- and *c*-lattice parameters, respectively. These results indicate that application of the SNH binder can suppress the structural evolution of the NCM811 cathode during cycling, and thereby yield stable cycling characteristics.

Comparison of ex-situ XRD patterns of PVDF electrode before and after cycling demonstrated that the peaks at (003) ($2\theta = 18.8^\circ$), (101) ($2\theta = 36.7^\circ$), and (108)/(110) ($2\theta = 64.5^\circ/65^\circ$), which correspond to the layered phase, were significantly changed after cycling tests (Figure 5D, Supporting Information).^[46] In contrast, in patterns of the SNH electrode, these peaks did not shift. These results demonstrate that SNH binder effectively prevents interfacial side reactions and dissolution of TM, and thereby yields excellent electrochemical stability.

To observe microscopic structural changes, we confirmed cross-sectional SEM images of SNH and PVDF electrodes after cycling tests (Figure 6A). The morphology of the NCM811 particles in the SNH electrode perfectly maintained their initial structure with tight covering by the binder, whereas the PVDF electrode was highly exposed and cracked without covering by the binder.

To gain a better understanding of the crystalline structural changes of NCM811 particles in both types of electrodes after cycling tests, transmission electron microscope (TEM) images were obtained (Figure 6B). The cycled NCM811 particles in the PVDF electrode changed phase and amorphous region. Ni²⁺ ions can migrate from TM slabs to Li slabs, where reactions yield a phase transformation from layered structure to the spinel structure, and to the formation of rock-salt structure.^[47] In contrast, cycled NCM811 particles in the SNH electrode maintained their

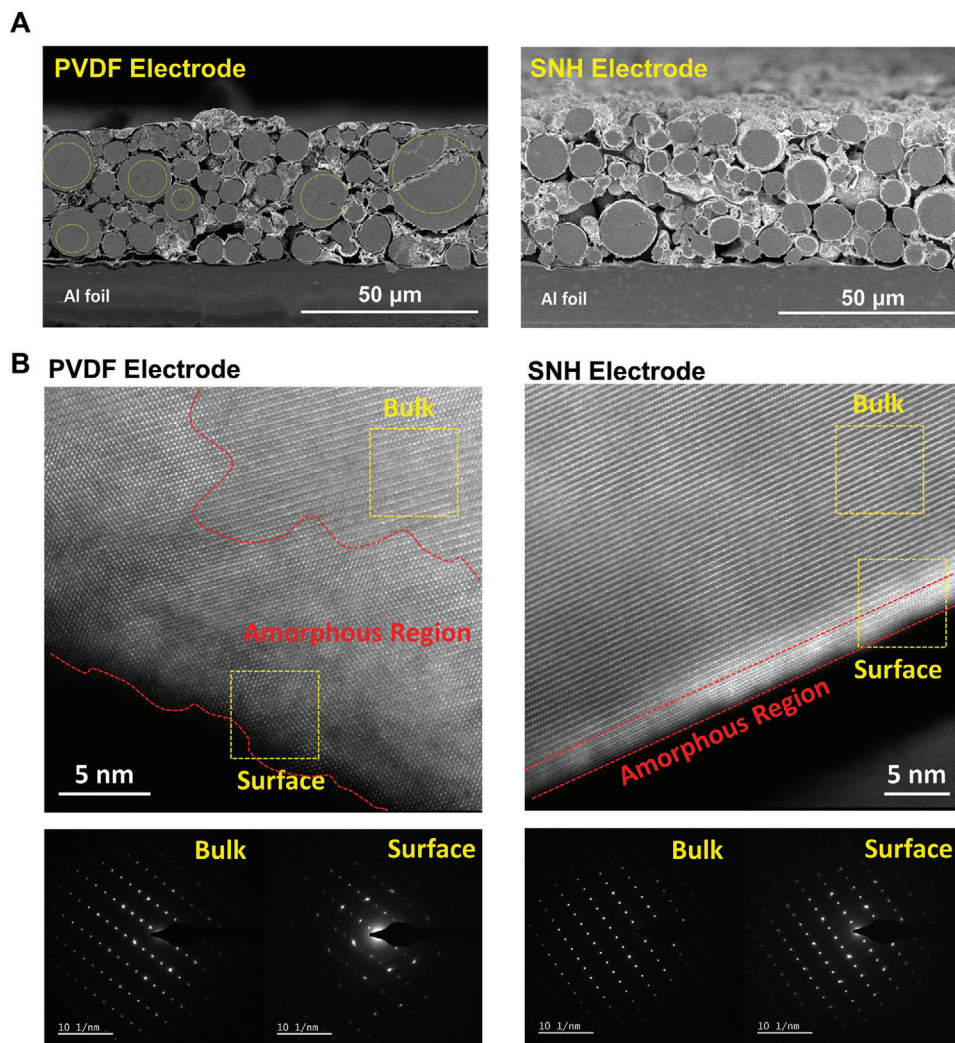


Figure 6. Morphology changes of SNH and PVDF electrodes after charge/discharge tests. A) cross-sectional SEM, B) TEM.

layered phase; a very thin rock-salt phase (< 5 nm) formed. These results support the results of previous elemental analyses.

2.5. Effect of SNH Binder in Full-Cell System of NCM811||Graphite

To verify the applicability of the SNH binder from a commercial perspective, the NCM811 electrode was investigated in a full-cell system that used a graphite anode that had been manufactured as described in the experimental section. The initial Coulombic efficiencies of both binders-applied electrodes were almost the same (SNH electrode: 98.46%; PVDF electrode: 97.89%), and were similar to those obtained in the half-cell tests. After 200 charge/discharge cycles, both types of electrodes showed almost retained Coulombic efficiency (SNH electrode: 100% (Figure 7A); PVDF electrode: 99.66% (Figure 7B). However, the SNH electrode full cell retained 87.82% of its initial specific capacity, whereas the PVDF electrode full cell retained only 61.24% (Figure 7C). TM elements (i.e., Ni, Mn and Co) in graphite an-

ode were less concentrated in the SNH electrode full cell (Ni = 78.9 ppb, Co = 0.38 ppb, Mn = 4.97 ppb) than in the PVDF electrode full cell (Ni = 110.2 ppb, Co = 8.0 ppb, Mn = 15.9 ppb) from ICP-MASS analysis after test (Figure S14, Supporting Information), and the graphite anode retained its layered structure (Figure S15, Supporting Information) from SEM analysis after test; these observations are further evidence for the excellent cyclability of SNH.

We propose a three-point mechanism by which SNH binder affects the electrochemical stability of NCM811-based LIBs. 1) SNH with F-free organic-inorganic nanohybrid structure alleviates pulverization of NCM811 particles. 2) The high polar silanol content in SNH induces strong interaction between SNH and NCM811 particles, which mitigates the lattice reconstruction of NCM811 particles. 3) Highly condensed inorganic network structure effectively prevents TM dissolution in NCM811 particles and thus restrains electrolyte attack, which induces the electrochemical stability of LIBs (Figure 8A). In contrast, PVDF with an F-bearing linear structure provides weak binder-cathode interaction and allows pulverization of NCM811 particles, which hugely

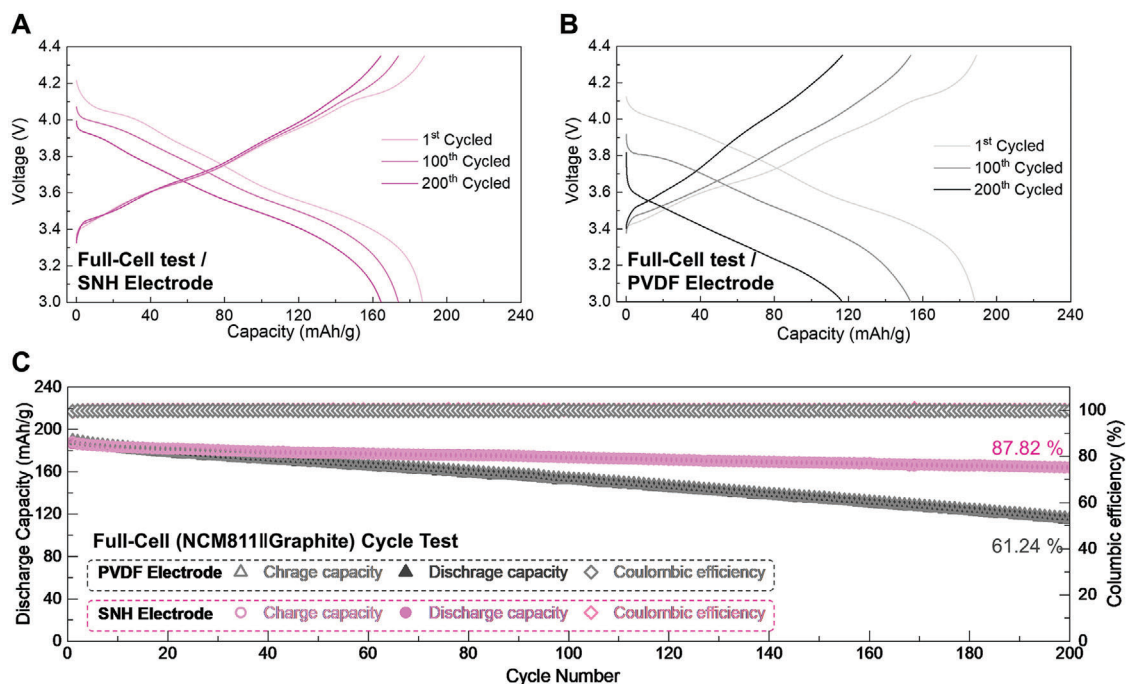


Figure 7. Cycle performance of LIB full cells using SNH and PVDF electrodes. A-B) Charge/discharge curves and C) capacity decay during cycling.

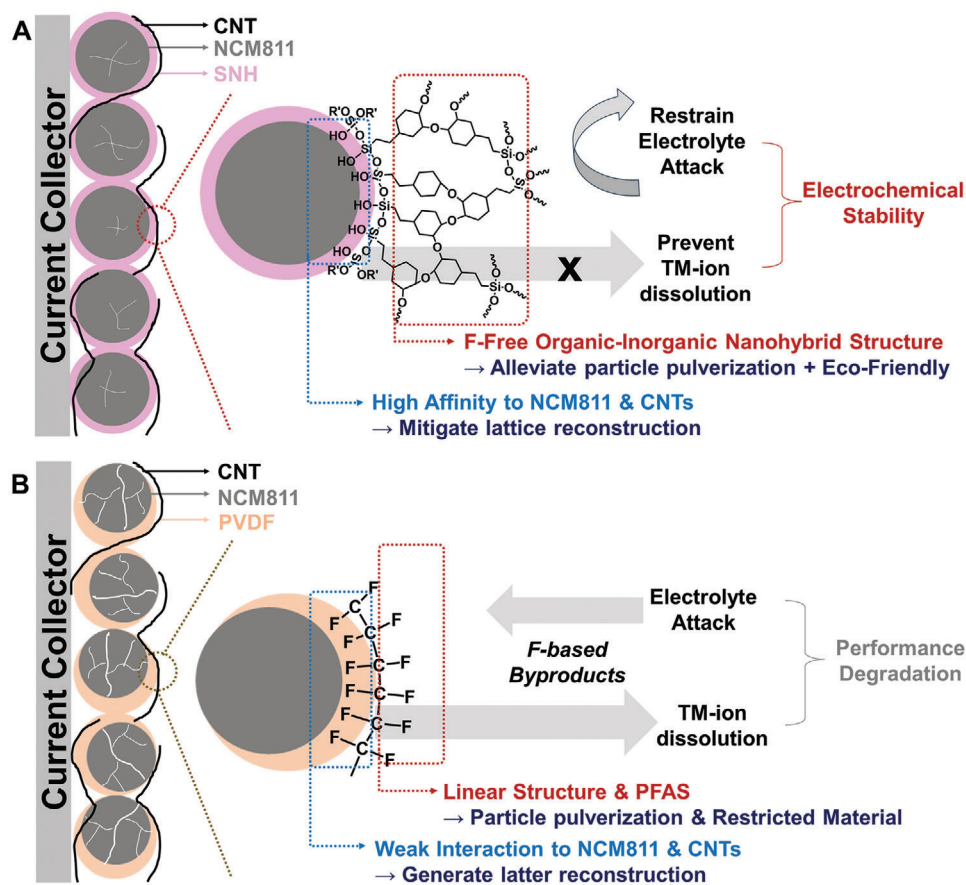


Figure 8. Schematic illustration of degradation mechanism of A) SNH electrode and B) PVDF electrode.

degrades the electrochemical characteristics of PVDF-based LIBs (Figure 8B).

3. Conclusion

In conclusion, we developed a novel binder-design strategy to enhance the cyclability of LIBs with high-capacity Ni-rich cathodes. This approach uses a simple sol-gel reaction of two silane precursors (ECTS and TEOS) to produce an SNH binder. The silanol (Si-OH) groups significantly improve the affinity for active materials and conductive agents, facilitating the formation of uniform electron/ion conducting pathways. Additionally, the chemical hybridization between organic functional groups (cycloaliphatic epoxy and silanol) and inorganic siloxane (Si-O-Si) networks enhances shear thinning, adhesion, and cohesion. Consequently, cathodes using the SNH binder show significantly improved capacity retention after cycling ($\approx 81.9\%$ after 200 cycles) compared to those using the PVDF binder ($\approx 58.8\%$ after 200 cycles). This improvement is attributed to the fluorine-free molecular structure and siloxane backbone rich in silanol, which effectively inhibits irreversible structural evolution in NCM811 and ensures the electrochemical stability of LIBs. The industrial effectiveness of the SNH binder was validated in a full-cell system. We believe that the development of this innovative fluorine-free siloxane nanohybrid-based binder is a highly effective approach for developing high-performance LIBs with high energy density and stable cycle life.

4. Experimental Section

Materials: PVDF (Kureha Co., Ltd., Japan), carbon nanotubes (CNTs) (dispersion, solution in NMP with hydrogenated nitrile rubber (HNBR, 20 wt% vs CNTs), Advanced Nano Products Co., Ltd., Korea), NMP (Daejung Chemical & Materials Co., Ltd., Korea), NCM811 particles (KY181, LG Chem., Korea), ECTS (Tokyo Chemical Institute, Japan), TEOS (Sigma-Aldrich, USA), ammonium hydroxide solution (28%, Sigma-Aldrich, USA), and cyclohexyl p-toluenesulfonate (Sigma-Aldrich, USA) were used without further purification.

Synthesis of Siloxane-Based Nanohybrid (SNH) Binder: ECTS and TEOS (molar ratio of ECTS to TEOS = 8 to 1) were put into a two-neck flask filled with 50% of NMP solvent and vigorously mixed using a magnetic stirrer at room temperature. Then ammonium hydroxide solution (28%) (0.11 phr of total silane weight) was added dropwise to the mixed solution while temperature was raised to 60 °C. Stirring for 8 h yielded a highly viscous and clear binder resin (SNH). Then 1 phr of cyclohexyl p-toluenesulfonate as a thermal initiator was added to the SNH to initiate ring-opening polymerization of epoxy groups.

Characterization: The FT-IR spectrum of SNH was obtained using an FT-IR 460 plus (JASCO, Japan). The ^{29}Si -NMR spectrum of SNH was recorded using a DMX600 FT 600 MHz (Bruker Biospin, Australia). Raman maps were obtained using LABRAM HR A (HORIBA, Japan). LSV curves were obtained by using WonA Tech, Korea in the voltage range of 2.5–5 V at a scan rate of 0.1 mV s $^{-1}$. The viscosity of slurries was recorded using a rheometer (Anton Paar Physica MCR 302) by using a parallel plate with a 25-mm diameter and 0.25-mm gap, with room-temperature frequency sweeps with decreasing frequency from 100 to 0.1 rad s $^{-1}$ (strain 0.5%). Adhesion and cohesion were quantified using LS1 (AMETEK, USA). To conduct adhesion and cohesion tests, both binders-applied slurry were coated onto Al foil and dried at 120 °C under vacuum overnight. Slide glass and each sample (electrode and binder resin) with a width of 20 mm were attached to 3 M double-sided tape and then peeled off at a 90° angle. After the adhesion test, 3 M adhesive tape with a width of 18 mm was applied to the cathode. The cohesion test was performed by a 90° peel test. Both adhesion and cohesion tests were conducted at a constant displacement rate

of 300 mm min $^{-1}$. SEM images were observed using a JSM-6700F (JEOL) under an accelerating voltage of 15 kV. X-ray CT images of slurry were obtained by 3D X-ray CT Scanner 450 (XTH-450, Nikon Metrology, U.S.A.). XPS spectra of freestanding electrodes were obtained using a Theta Probe AR-XPS System (Busan Center, KBSI, Korea) with ultrahigh-vacuum chambers. TOF-SIMS surface analysis was performed using TOF-SIMS5 (ION TOF GmbH, Germany) to confirm the surface analysis of both binders-applied electrodes. A pulsed Bi $^{3+}$ ion beam (30 keV) set in high current mode was used for TOF-SIMS surface analysis. *Operando* XRD was measured using an X-ray diffractometer (Empyrean, PANalytical) with Mo K α radiation ($\lambda = 0.709 \text{ \AA}$). The XRD patterns were collected in the range $7 \leq 2\theta \leq 23^\circ$ with a step size 0.016° during charge/discharge at a current density of 30 mA g $^{-1}$ within the operating voltage range $3 \leq V_{\text{OP}} \leq 4.35 \text{ V}$ (vs Li $^+$ /Li). *Operando* XRD patterns were obtained to monitor structural change during charge/discharge at a current density of 30 mA g $^{-1}$ within the range $3 \leq V_{\text{OP}} \leq 4.5 \text{ V}$ (vs Li/Li $^+$). *Operando* XRD patterns of electrodes to which both binders had been applied were also measured using an X-ray diffractometer (XRD, PANalytical) with a step size 0.013° , which was transformed using Cu K α radiation ($\lambda = 1.54178 \text{ \AA}$). Ex situ XRD patterns were characterized using X-ray diffraction (XRD, PANalytical) with Mo K α radiation ($\lambda = 0.70932 \text{ \AA}$). The XRD patterns' angles were transformed using Cu K α radiation ($\lambda = 1.54178 \text{ \AA}$) to enable comparison with previous research. STEM images were observed by TEM using a JEOL JEM-F200, operated at 300 keV. GITT tests were performed at 0.5C rate under current pulse for 300 sec. in the voltage range of 3–4.35 V. Thermal properties of both binders were examined via thermogravimetric analysis (TGA, Q50, TA Instruments, Inc.) with a 10 °C min $^{-1}$ ramp under a N $_2$ atmosphere.

Fabrications of Electrodes and Electrochemical Characterization: Slurries of electrode materials were obtained by mixing NCM811 (0.2 kg), CNT dispersion (0.8 wt% vs NCM811), and binder (1.2 wt% vs NCM811) solution, then applying high-shear mixing at 3000 rpm for 15 min. Solid contents of SNH in NMP solvents were 50 wt%. Electrode layers were prepared using the doctor-blade method on Al foils, then dried at 80 °C for 2 h, then at 120 °C for 12 h. The mass loading of the electrode layer was $\approx 10 \text{ mg-cm}^{-2}$ for cell tests.

A universal testing machine (UTM) to measure adhesiveness (refers to the force between the electrode and current collector) and cohesiveness (refers to the force between the electrode and electrode) strength was used for a 90° peeling test of electrodes under 25 mm min $^{-1}$. For adhesiveness measurements, pressing electrodes (mass loading: $\approx 12 \text{ mg cm}^{-2}$, density: 3 mg cm $^{-3}$ and area: 2 × 7 cm) with a length of 4 cm were immobilized to tape slide glass by using a double-sided adhesive. The sample for the adhesion test was prepared by attaching a double-sided tape on a sliding glass and then attaching a current collector (Al-foil)-electrode to it, and the size was 50 mm in width and 20 mm in depth (Figure S16A, Supporting Information). After loading the prepared sample onto the UTM, a 90° peel test was performed at a fixed speed of 300 mm min $^{-1}$ to calculate the adhesion value. The average force was calculated within the intermediate region (20 to 80% of the separation range). In the case of the cohesion test, a single-sided tape was attached to the sample where the adhesion test was completed, and a 90° peel test was performed to calculate the cohesion value (Figure S16B, Supporting Information).

To evaluate the electrochemical characteristics of the electrodes, coin-cells (2032 coin) were assembled in a glove box under highly pure Ar. The cells consisted of an electrode, a PP separator (Celgard 2400), an electrolyte (1 M LiPF $_6$ in EC:DMC (ethylene carbonate:dimethyl carbonate), 1:1 by volume), and a Li counter electrode. The punched electrodes in cells had areas of 1.538 cm 2 . Galvanostatic charge/discharge tests for NCM811 were performed at $3 \leq V_{\text{OP}} \leq 4.35 \text{ V}$ at 45 °C and C rates of 0.1, 0.33, 0.5, 1, 2, 3, 4 and 6C, where 1C = 230 mA g $^{-1}$.

Long-term cycling tests of coin-cell/full-cell were performed at $3 \leq V_{\text{OP}} \leq 4.35 \text{ V}$ at 45 °C and 0.5C/1C as follows. First, two cycles at 0.1C/0.1C, three or four cycles at 0.33C/0.33C then five or size cycles at 0.5C/0.5C. Charge/discharge curves were recorded using a WBCS 3000 battery tester system (WonA Tech). Electrochemical impedance spectroscopic analysis was performed at frequencies of 0.1 MHz to 0.1 Hz with a voltage amplitude of 0.01 V, using a ZIVE SP1 electrochemical workstation (WonA Tech). Graphite anodes for full cells were composed of graphite (91%),

carbon black (1%), and Carboxymethyl cellulose (CMC)/styrene-butadiene rubber (SBR) (8%). Prior to the assembly of NCM811/Graphite full cells, the graphite anodes were assembled in half cells and subjected to two charge-discharge cycles to reduce irreversible capacity decay in the full-cell tests. The N/P ratios of the NCM811 cathodes and graphite anodes were ≈ 1.3 . Each full cell was composed of an NCM811 cathode, a PP separator (Celgard 2400), an electrolyte (1 M LiPF₆ in EC:DMC (1:1, v/v%), and a graphite anode. The electrode for the full-cell was designed with an N/P ratio of 1.3, which is higher than the conventional N/P ratio value of 1.11 to 1.12, in order to minimize the influence of the anode.

Computational Details: DFT calculations were performed using the Gaussian 16 software package.^[48] Projector-augmented wave pseudopotentials were employed with a plane-wave basis set, as implemented in VASP.^[49] Perdew–Burke–Ernzerhof's (PBE) parameterization of the generalized gradient approximation (GGA) was utilized.^[50] A $2 \times 2 \times 1$ k-point grid was applied for DFT calculations to model the NCM811 surface. To achieve more accurate results regarding the van der Waals interactions between the SNH (or PVDF) binder and the surface, the DFT-D3 correction method was incorporated in this study.^[51] The GGA+U method^[52] was adopted to address the localization of d-orbitals in Ni ($U = 6.0$ eV), Co ($U = 3.4$ eV), and Mn ($U = 3.9$ eV), which is consistent with the previous report.^[53] A kinetic energy cutoff of 500 eV was used in all calculations, and the overall structure was optimized until the force within the unit cell converged to within $0.03 \text{ eV } \text{\AA}^{-1}$.

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

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Conflict of Interest

The authors declare no conflict of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Keywords

binder, fluorine-free, lithium-ion batteries, Ni-rich layered cathode, siloxane nanohybrid

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